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Antimycotic Potency of Some Essential Oils in the Treatment of Induced Dermatomycosis of an Experimental Animal

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Abstract

During screening of 20 essential oils against *Epidermophyton floccosum* and *Microsporum gypseum*, the test dermatophytes, the oils of *Ocimum gratissimum* and *Trachyspermum ammi* exhibited strong antidermatophytic properties. The ointments of the oils blocked dermatomycosis induced with guinea pigs, thereby indicating their efficacy as herbal treatments for the cure of dermatomycoses.

Keywords: Antimycotic, dermatomycoses, essential oils, fungitoxicity, guinea pigs, *Tinea*.

Introduction

The dermatophytes are a group of fungi that have the ability to invade the cutaneous tissues of animals and human beings. The lesions produced by such fungi are limited to hair, nails and the stratum corneum of the skin. These dermatophytic infections may involve small to significant areas of the cutaneous tissues while producing from mild cosmetic problems to debilitations and disfiguring diseases. Dermatomycoses are among the most common infectious diseases in animals and humans. During the latter half of this century, some effective synthetic antifungal drugs have been discovered for the treatment of dermatomycoses. Due to their various side effects on human systems, the use of most such drugs has remained limited (Peterson et al., 1980; Symons et al., 1980; Wajtulewski et al., 1980). Amphotericin-B and nystatin, which are commonly used as chemotherapeutic agents, also have some side effects (Pareek, 1980).

Higher plants and their products have been used in different traditional medicines since time immemorial. Because of their long history of use in the treatment of different human ailments, most of the herbal drugs are believed to be safer than the synthetic drugs and there would be much

less chance of side effects after application. Higher plants remain as an almost untapped reservoir of potentially useful chemical compounds not only as drugs, but also as unique templates that could serve as a starting point for synthetic analogues and as interesting tools that can be applied to a better understanding of biological processes. India ranks foremost after South Korea in the supply of medicinal plants to industrialised countries where demands for natural drugs have been increasing in recent years (Chadha et al., 1980). Therefore, in the present study, the potential of some essential oils of higher plants against fungi causing dermatomycosis was determined. *Epidermophyton floccosum* (Harz) Langeron & Miloch., and *Microsporum gypseum* (E. Bodin) Guiart & Grigoraki, which are both anamorphic Ascomycetes, cause severe dermatomycoses as *Tinea corporis*, *Tinea pedis* and onychomycosis, and have been used here as the test dermatophytes.

Materials and methods

Higher plants of different angiospermic families of the local region were randomly selected and identified. Confirmation of the identity of the plants was done with the help of authentic herbarium specimens deposited in the Herbarium of the National Botanical Research Institute, Lucknow, India.

Essential oils of these plants were tested for their fungitoxicity against test fungi by the poisoned food techniques. Fresh parts of each plant (250g) were cut separately into small pieces and then thoroughly washed with sterilized water. The volatile fractions (essential oils) were isolated by hydrodistillation using a Clevenger apparatus. In case of essential oil bearing plants, the collecting funnel of the Clevenger apparatus showed two distinct layers – an upper

oily layer and lower aqueous layer. The layers were separated and the essential oils were stored in clean glass vials after removing water traces with the help of capillary tubes and anhydrous sodium sulphate.

The fungitoxicity of the essential oils was tested by the poisoned food techniques of Perrucci et al. (1994). Sabouraud dextrose agar medium (SDA) was prepared, autoclaved and cooled to $40 \pm 2^\circ\text{C}$. Requisite amounts of the oils were dissolved separately in 0.5 ml of a 0.01% aqueous solution of Tween-80 in pre-sterilized Petri plates (7.5 cm). SDA medium (9.5 ml) was pipetted into each Petri plate and mixed thoroughly so as to obtain the required concentrations of 500 and 1000 ppm. For control sets the necessary amount of sterilized water in place of the oil was added to the medium containing 0.5 ml of Tween-80. The experiment was run in triplicate and repeated twice. Discs of the test dermatophytes, *Epidermophyton floccosum* and *Microsporum gypseum* (5 mm diameter), were cut from the periphery of a 7-day-old culture with the help of a sterilized cork borer and were aseptically inoculated upside down to the center of each Petri plate of treatment and control sets. The Petri plates were incubated at $28 \pm 1^\circ\text{C}$ for six days in an incubation chamber. Diameters of fungal colonies of treatment and control sets were measured in mutually perpendicular directions on the 7th day. The percentage of inhibition of growth of test dermatophytes was calculated by the mean value of colony diameters by the following formula:

Percent inhibition of mycelial growth

$$= [(dc - dt)/dt] \times 100\%$$

where dc = average fungal colony diameter in control sets; dt = average fungal colony diameter in treatment sets.

To find the minimum inhibitory concentration at which the oil showed absolute fungitoxicity, experiments were carried out by the usual poisoned food technique. Different concentrations of the oils: 25, 50, 100, 200, 300, 400, and 500 ppm, were prepared by dissolving separately their requisite amount in 0.5 ml of 0.01% aqueous Tween-80 and then mixing with 9.5 ml Sabouraud dextrose agar medium. The medium of control sets contained the requisite amount of sterilised water dissolved in 0.5 ml Tween-80 in place of the oils. As usual the prepared plates were inoculated aseptically with the assay disc of the test dermatophytes and incubated for 6 days at $25 \pm 2^\circ\text{C}$. The observation was recorded on the seventh day and the percent inhibition of growth of the test dermatophyte was calculated.

The effect of increased inoculum density of both of the test dermatophytes on the fungitoxicity of the essential oils of *Ocimum gratissimum* L. (Lamiaceae) and *Trachyspermum ammi* (L.) Sprague ex Turril (Apiaceae) was studied following Venkatramana and Pattisapu (1987). Requisite quantities of the oils were dissolved in 0.5 ml of 0.01% aqueous Tween-80 and then mixed to 9.5 ml liquid Sabouraud dextrose agar medium to make the final concentration of 50 ppm for *Ocimum gratissimum* and 200 ppm for *Trachyspermum ammi* oil. Seven sets thus prepared were inoculated separately by

the assay discs (4 mm in diameter) of each test dermatophyte in a geometrical progression of two, i.e. 1, 2, 4, 8, 16, 32 and 64. The approximate number of spores of each set was recorded with the help of a haemocytometer by introducing the similar number of fungal discs in distilled water. For the control set an equal quantity of sterilized water dissolved in Tween-80 was mixed with the liquid Sabouraud dextrose agar medium. All the sets were incubated for six days at $25 \pm 2^\circ\text{C}$ and observations were recorded on the 7th day.

The antifungal spectrum of the oils was determined at their respective minimum inhibitory concentrations, that is, at 50 ppm for *Ocimum gratissimum* and 200 ppm for *Trachyspermum ammi* by the usual poisoned food technique using Sabouraud dextrose agar medium against 12 human pathogenic fungi: *Aspergillus flavus* Link:Fr., anamorphic Trichocomaceae; *A. niger* Tiegh., *Candida albicans* (C.P. Robin) Berkhout, anamorphic Saccharomycetales; *Cryptococcus neoformans* (San Felice) Vuill., anamorphic Filobasidiaceae; *Epidermophyton floccosum* (Harz) Langeron & Miloch., anamorphic Hyphomycetes; *Malassezia pachydermata* (Weidman) Dodge, anamorphic Hyphomycetes; *Microsporum gypseum* (E. Bodin) Guiart & Grigoraki, anamorphic Onygenales; *M. audouinii* Gruby; *M. nanum* C.A. Fuentes; *Scopulariopsis brevicaulis* (Sacc.) Bainier, anamorphic Microascaceae; *Trichophyton rubrum* (Castell.) Sabour., anamorphic Arthrodermataceae; and *T. violaceum* Sabour. ex E. Bodin.

The observations were recorded and percentage inhibition of growth of fungi was calculated. The nature of fungitoxicity (fungistatic/fungicidal) was also observed against absolutely sensitive fungi by the standard techniques of Garber and Houston (1959) and Thompson (1989).

The essential oils of *Ocimum gratissimum* and *Trachyspermum ammi* were tested for their therapeutic potency as antimycotic agents in the treatment of induced dermatomycoses of guinea pigs. Guinea pigs (weighing 350 to 400 g, age 5 to 8 months) were inoculated separately by *Microsporum gypseum* and *Epidermophyton floccosum* by the method of Srivastava et al. (1978). The hair of an area of approximately 9 cm^2 on either side of central line of the back of the animals was removed with clippers and then shaved with a safety razor. An abrasion in the shape of a cross was made on the area shaved with the help of a sterilized scalpel. The test dermatophytes were grown separately on Sabouraud dextrose agar slants at $28 \pm 1^\circ\text{C}$ for 10 days for inocula. The mycelial mat removed from each slant was macerated separately with 500 mg sterilized white sand (SiO_2) and a paste was made with 2.5 ml of autoclaved and cooled honey. The paste was applied on the cleaned area and was abraded on skin with no.1 grit sand paper. The freshly inoculated area of the guinea pig was covered with a piece of sterilized plastic sheet with the help of adhesive tape for 2 h. A set of 8 animals was used for each treatment and control set. After 2 days post-inoculation, that is, on the third day, isolation of dermatophytes from the infected area was done to ascertain the establishment of infection following the method of Moore

and Jaciow (1979). The infected areas were cleaned with 70% ethanol. The scales and hair stubs collected from the active border, that is, the outer edge of the infected area containing the viable organism, were placed in sterilized dry Petri plates for direct examination and confirmatory culture test. After confirmation of the establishment of infection further *in vivo* studies were made through the ointments prepared from the oils of the *Ocimum gratissimum* and *Trachyspermum ammi*. The ointments to contain each oil were prepared by mixing 2 concentrations of polyethylene glycol (1500 and 200) and the creamy base was then mixed well with the necessary amount of the oil to obtain a concentration of 1000 ppm. The treatment of infected animals was started from the sixth day after inoculation and continued until complete recovery from the infection was achieved. The ointment (about 400–500 mg) was applied twice a day – in the morning and in the evening. Cultures for the examination were made using scales and hair stubs collected from active border of the infection sites on alternate days by the usual method of Moore and Jaciow (1979). The results were recorded in terms of percent culture recovery following Wahab et al. (1982) by the formula as follows:

$$\text{Percent culture recovery} = (\text{Total no. of sites positive for culture of each set} / \text{Total no. of sites in each set}) \times 100\%$$

Results

It is evident from Table 1 that out of 20 essential oils tested against *Epidermophyton floccosum* and *Microsporum gypseum*, all the oils except for those of *Murraya koenigii* (L.) Spreng. (Rutaceae) and *Tagetes erecta* L. (Asteraceae) showed absolute fungitoxicity against both test dermatophytes at 1000 ppm. Even at 500 ppm most of the oils showed strong (>70%) fungitoxicity. However, the oils of *Ocimum gratissimum* and *Trachyspermum ammi* were completely inhibitory to *Epidermophyton floccosum* at 500 ppm. Similarly, the oils of *Artemisia vulgaris* L. (Asteraceae) *Ocimum gratissimum* and *Trachyspermum ammi* showed absolute inhibition of growth of *Microsporum gypseum*. Therefore, the oils of *Ocimum gratissimum* and *Trachyspermum ammi* were selected for further studies in the present investigation.

It is evident from Table 2 that *Ocimum gratissimum* oil completely inhibited the growth of the test dermatophytes at 50 ppm while the oil of *Trachyspermum ammi* was completely inhibitory at 200 ppm. These concentrations were therefore recorded as the minimum inhibitory concentrations. The oils of *Ocimum gratissimum* and *Trachyspermum ammi* were in fact altogether fungicidal in nature even at their MIC of 50 and 200 ppm, respectively.

Table 1. Screening of some essential oils against dermatophytes.

Essential Oil	Percent Inhibition of Growth			
	<i>Epidermophyton floccosum</i>		<i>Microsporum gypseum</i>	
	500 ppm	1000 ppm	500 ppm	1000 ppm
<i>Aegle marmelos</i> (L.) Correa, Rutaceae	90	100	85	100
<i>Artemisia vulgaris</i> L., Asteraceae	95	100	100	100
<i>Caesulia axillaris</i> Roxb., Asteraceae	95	100	80	100
<i>Cinnamomum tamala</i> (Buch.-Ham.) Nees & Eberm., Lauraceae	90	100	95	100
<i>Cinnamomum camphora</i> (L.) J. Presl, Lauraceae	85	100	85	100
<i>Citrus medica</i> L., Rutaceae	85	100	72	100
<i>Corymbia citriodora</i> (Hook.) K.D. Hill & L.A.S. Johnson, Myrtaceae	80	100	75	100
<i>Eupatorium cannabinum</i> L., Asteraceae	85	100	80	100
<i>Lippia alba</i> (Mill.) N.E. Br. ex Britton & P. Wilson, Verbenaceae	80	100	90	100
<i>Mentha arvensis</i> L., Lamiaceae	95	100	90	100
<i>Murraya koenigii</i> (L.) Spreng., Rutaceae	70	95	70	95
<i>Nepeta hindostana</i> (Roth) Haines, Lamiaceae	85	100	85	100
<i>Ocimum gratissimum</i> L., Lamiaceae	100	100	100	100
<i>Citrus sinensis</i> (L.) Osbeck, Rutaceae	60	80	55	80
<i>Tagetes erecta</i> L., Asteraceae	80	80	70	75
<i>Trachyspermum ammi</i> (L.) Sprague ex Turrill, Apiaceae	100	100	100	100
<i>Vetiveria zizanioides</i> (L.) Nash, Poaceae	80	100	75	100
<i>Vitex negundo</i> L., Lamiaceae	60	100	75	100
<i>Zanthoxylum acanthopodium</i> D.C., Rutaceae	95	100	90	100
<i>Zingiber officinale</i> Roscoe, Zingiberaceae	80	100	85	100

Table 2. Minimum inhibitory concentration and nature of *Ocimum gratissimum* and *Trachyspermum ammi* oil.

Concentration in ppm	Percent inhibition of growth			
	<i>Ocimum gratissimum</i>		<i>Trachyspermum ammi</i>	
	<i>E. floccosum</i>	<i>M. gypseum</i>	<i>E. floccosum</i>	<i>M. gypseum</i>
500	100*	100*	100*	100*
400	100*	100*	100*	100*
300	100*	100*	100*	100*
200	100*	100*	100*	100*
100	100*	100*	80	80
50	100*	100*	50	50
25	60	50	35	30

* Denotes fungicidal nature.

Table 3. Efficacy of the oils at increased inoculum density of the test dermatophytes.

Oils	Growth of the test dermatophytes									
	<i>E. floccosum</i> with increased inoculum density					<i>M. gypseum</i> with increased inoculum density				
No. of Discs	2	4	16	32	64	2	4	16	32	64
<i>O. gratissimum</i>	–	–	–	–	+	–	–	–	–	–
<i>T. ammi</i>	–	–	–	–	–	–	–	–	–	–
Control	+	+	+	+	+	+	+	+	+	+

+ denotes growth; – denotes no growth.

The results provided in Table 3 show that the oils inhibited the growth of the test dermatophytes in treatment sets containing even 64 discs indicating their capacity to withstand high inoculum densities. The *Ocimum gratissimum* oil, however, could not inhibit 64 discs of *Epidermophyton floccosum*.

It is evident from Table 4 that the oils exhibited a broad antifungal spectrum. However, the nature of fungitoxicity of the oils varied with test fungi. *Ocimum gratissimum* oil was fungistatic against *Aspergillus flavus*, *A. niger* and *Trichophyton violaceum* while *Trachyspermum ammi* oil showed fungistatic activity against *A. niger* and *T. violaceum*. Both the oils were fungicidal against the rest of the fungi.

Tables 5 and 6 indicate that the ointments of the oils showed curative potency against induced dermatomycoses of guinea pigs. The ointment of the oil of *Ocimum gratissimum* could completely cure the induced dermatomycoses caused by *Epidermophyton floccosum* and *Microsporum gypseum* by 11 and 9 days, respectively. The ointment of the oil of *Trachyspermum ammi* oil could completely cure the infection induced by *Epidermophyton floccosum* and *Microsporum gypseum* by 13 and 11 days, respectively.

Discussion

Although the screening of plant extracts against fungi has been done previously, most researchers have used plant pathogenic fungi in their screening programs. In order to develop chemotherapeutic agents of plant origin, the initial step should be a proper screening of the local plants in order to select efficacious ones for further phytochemical and fungitoxic studies.

These oils were tested against 12 human pathogenic fungi and were found to exhibit a broad antidermatophytic spectrum inhibiting all the human pathogenic fungi tested. The broad antimicrobial efficacy of these oils indicates their larger market value if used as chemotherapeutic agents. During assessment of their antimicrobial spectrum, the nature of fungitoxicity of the oils was also found to vary with the organisms. Thus, a fungitoxic product may exhibit fungicidal activity against one microorganism but it may show fungistatic activity against another.

The efficacy of an antibiotic substance may also depend on the number of organisms it has to combat (Skinner, 1955). At a particular concentration, the potency of a fungitoxic compound can thus depend on the inoculum density of the

Table 4. Antidermatophytic spectrum of *Ocimum gratissimum* and *Trachyspermum ammi* oils.

Human Pathogens	Percent inhibition of growth	
	<i>Ocimum gratissimum</i> oil at 50 ppm	<i>Trachyspermum amii</i> oil at 200 ppm
<i>Aspergillus flavus</i> Link:Fr., anamorphic Trichocomaceae	100**	100*
<i>A. niger</i> Tiegh.	100**	100**
<i>Candida albicans</i> (C.P. Robin) Berkhout, anamorphic Saccharomycetales	100*	100*
<i>Cryptococcus neoformans</i> (San Felice) Vuill., anamorphic Filobasidiaceae	100*	100*
<i>Epidermophyton floccosum</i> (Harz) Langeron & Miloch., anamorphic Hyphomycetes	100*	100*
<i>Malassezia pachidermata</i> (Weidman) Dodge, anamorphic Hyphomycetes	100*	100*
<i>Microsporum gypseum</i> (E. Bodin) Guiart & Grigoraki, anamorphic Onygenales	100*	100*
<i>M. audouinii</i> Gruby	100*	100*
<i>M. nannum</i> C.A. Fuentes	100*	100*
<i>Scopulariopsis brevicaulis</i> (Sacc.) Bainier, anamorphic Microascaceae	100*	100*
<i>Trichophyton rubrum</i> (Castell.) Sabour., anamorphic Arthrodermataceae	100*	100*
<i>T. violaceum</i> Sabour. ex E. Bodin	100**	100*

* denotes fungicidal, ** denotes fungistatic.

Table 5. Efficacy of the ointments of *Ocimum gratissimum* and *Trachyspermum ammi* oils in cure of dermatomycosis on guinea pigs induced by *Epidermophyton floccosum*.

Treatment days	Percent culture recovery		
	Control	<i>O. gratissimum</i>	<i>T. ammi</i>
3	100	100	100
5	100	100	100
7	100	100	100
9	100	50	100
11	100	0	70
13	100	0	0
15	100	0	0

Table 6. Efficacy of the ointments of *Ocimum gratissimum* and *Trachyspermum ammi* oils in cure of dermatomycosis on guinea pigs induced by *Microsporum gypseum*.

Treatment days	Percent culture recovery		
	Control	<i>O. gratissimum</i>	<i>T. ammi</i>
3	100	100	100
5	100	100	100
7	100	50	100
9	100	0	65
11	100	0	0
13	100	0	0
15	100	0	0

test fungus. For example, the antifungal activities of citral and menthol were found to vary with inoculum density (Venkataraman & Pattisapu, 1987). This suggests that a high inoculum density of a fungus may either degrade or convert the compound into detoxified forms. The nature of detoxification of the fungitoxic products will also depend on the detoxifying enzymes produced by them. Thus, the detoxification of fungitoxic products at varying inoculum densities could be one of the main factors limiting their efficiency as ideal antifungal agents, leading to the minimum inhibitory concentration of the antifungal compound depending on inoculum density. For example, the increased inoculum density of fungi has been shown to reduce the

antifungal potency of *Allium sativum* L. (Alliaceae) (Misra 1975). On the other hand, in the case of the essential oils of *Cedrus deodara* (Roxb. ex D. Don) G. Don (Pinaceae) (Dikshit, 1980), and *Peperomia pellucida* K.S. Kunth (Piperaceae) (Singh et al., 1983), an increased inoculum density did not affect their antifungal potency. In the present study the essential oils of *Trachyspermum ammi* and *Ocimum gratissimum* could inhibit the highest inoculum doses used (up to 64 fungal discs) of the some dermatophytes, even at their MICs, with the exception of the essential oil of *Ocimum gratissimum*, where the revival of growth of *E. floccosum* was noticed on introduction of 64 discs.

The potency of the essential oils of *Ocimum gratissimum* and *Trachyspermum ammi* in the treatment of dermatomycosis was observed in guinea pigs by inoculating cultures of *Microsporum gypseum* and *Epidermophyton floccosum* separately. The ointments of the oils were prepared in polyethylene glycol (PEG) as it was reported to be a skin penetration enhancer and to increase the skin permeability (Osborne & Henke, 1997). Besides, PEG includes water solubility and compatability with many dermatological medicaments (Robinson et al., 1964). The application of the ointment of the essential oil of *Ocimum gratissimum* in PEG was shown to cure the experimentally induced dermatomycoses caused by *Epidermophyton floccosum* and *Microsporum gypseum* in 11 and 9 days, respectively, while the ointment of *Trachyspermum ammi* could cure the infections by the respective dermatophytes in 13 and 11 days. The absence of the propagules of the dermatophytes during percent culture recovery tests was indicative of the chemotherapeutic activity of the ointments of oils. Guinea pigs were used as test animals as they have been reported to be highly susceptible animals for dermatomycosis (Nandi & Bose, 1976).

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